Research Summary Sheet - Article 5

Complete the following form for each artifact that validates your problem statement.

Source (APA format and include a link to the source)

Mckee, P. (2025). *A survey of autonomous navigation techniques applicable to lunar surface exploration*.

https://ntrs.nasa.gov/api/citations/20250000993/downloads/AAS_25_173_SurfaceNav_Presentation_draft012425.pdf

Artifact Summary

This article examines the technologies for lunar rovers to operate autonomously. It studies sensors such as LiDAR, various cameras, IMUs, wheel encoders, and star trackers, as well as hazard detection and pathfinding. The study also covers navigation/communication systems when Earth is not in reach or unavailable. This connects to my HUNCH project because the robot must safely be able to navigate from the lander to the habitat without any human control.

Artifact Critique

This source is relevant because it explains many useful technologies to solve any navigation issues that could arise. Furthermore, we can build off of this to design our robot to operate autonomously. For my HUNCH project, the strongest point of evidence is the comparison of different sensors and navigation methods. However, it isn't structured like an experiment and rather like a survey, which help gather data but not with implementing it.

Reliability

McKee is a researcher in NASA Johnson Space Center's Aeroscience and Flight Mechanics Division. His work was presented at the 47th American Astronautical Society Guidance, Navigation, and Control Conference in 2025. It's a strong reference due to its intricate explanation of the techniques and NASA research.